

APS110 Tutorial #4

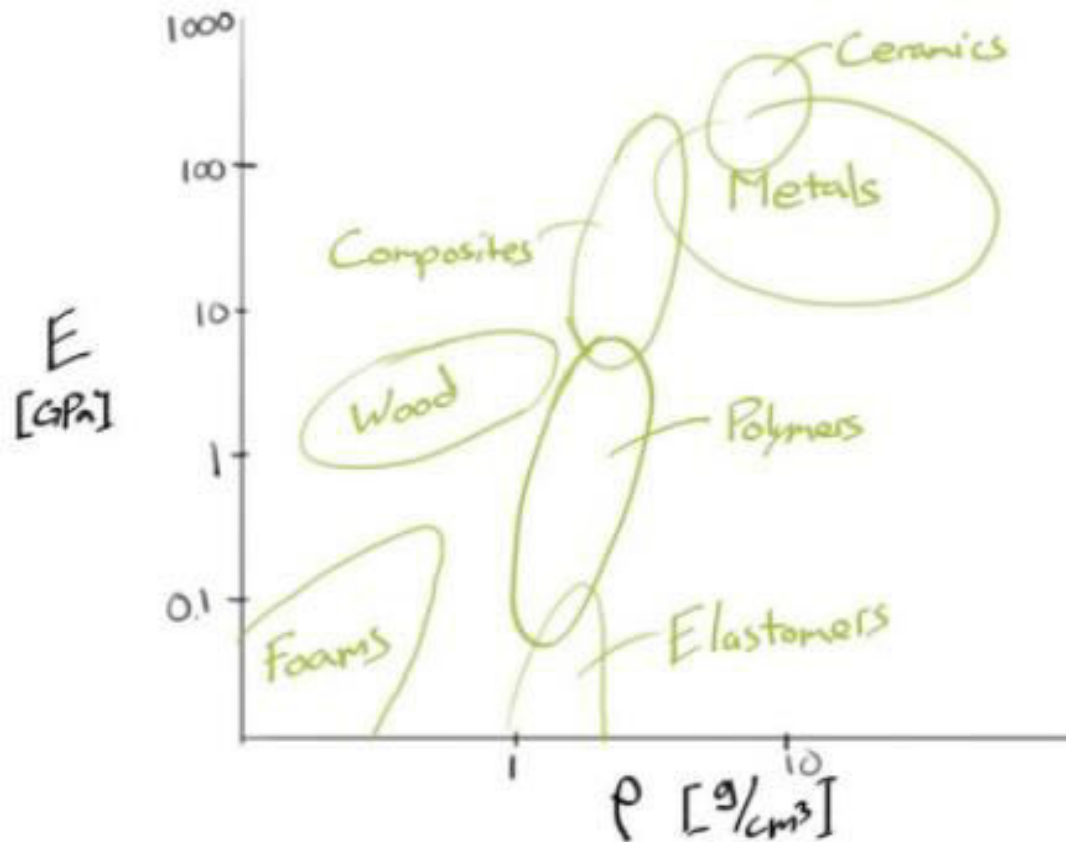
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Announcements

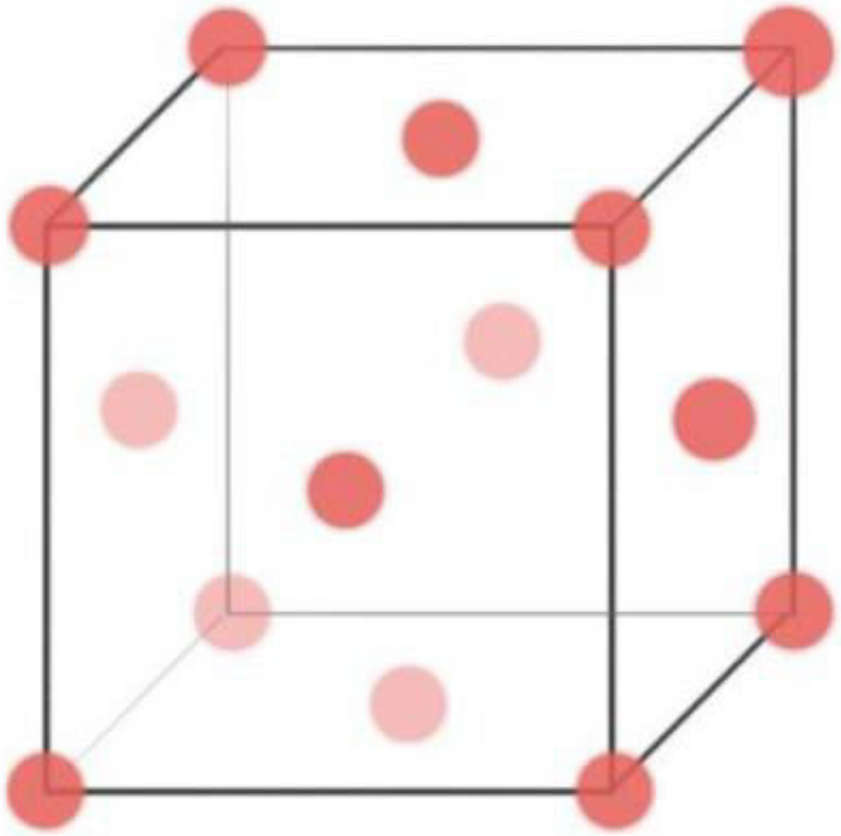
1. How did the first quiz go? Issues? Things that could be improved upon? Was it enough time?
2. First problem set will cover Chapters 1 through 4 and is posted. It's due by end of day on October 15th and should be submitted on Quercus as a file upload – pdf preferred – and it must be a single file.

Review



- Materials that are resistant to deformation are also more dense
- Remembering the attraction/repulsion force graph, E is related to the repulsive interactions between atoms
- Materials with more atoms per unit volume will have more repulsive interactions occur when they are deformed, and so will resist deformation better

Review



- Coordination number is the number of nearest neighbour atoms
- In the FCC structure, the CN is 12
- The atomic packing factor for FCC is 0.74

- $APF = \frac{V_{atoms}}{V_{cell}} = \frac{n \cdot \frac{4}{3}\pi R^3}{a^3}$
- $a_{FCC} = 2\sqrt{2}R$
- $n_{FCC} = 4$

Practice Question 1

1. Quiz Review

Practice Question 2

1. Platinum forms the FCC crystal structure and has a density of 21.5 g/cm³. Calculate the value of the lattice parameter in Å. ($A_{\text{Pt}} = 195.08 \text{ g/mol}$)

$$n_{\text{FCC}} = 4$$

$$A_{\text{Pt}} = 195.08 \text{ g/mol}$$

$$\rho = 21.5 \text{ g/cc} = 21.5 \times 10^6 \text{ g/m}^3$$

$$\rho = \frac{nA}{V_c N_A}, V_c = a^3 \Rightarrow a = \left(\frac{nA}{\rho N_A} \right)^{\frac{1}{3}}$$

$$a = \left(\frac{4 \cdot 195.08}{(21.5 \times 10^6) \cdot (6.022 \times 10^{23})} \right)^{\frac{1}{3}}$$

$$a = 3.92 \times 10^{-10} \text{ m} = 3.92 \text{ Å}$$

Practice Quiz

reasons:

- a) Because Young's modulus relates to a specific bond type, which does not change with microstructural changes
 - b) Because Young's modulus is a dimensionless quantity
 - c) Because Young's modulus does not depend on the size of a sample being tested
 - d) None of the above
2. The stored elastic strain energy in tempered glass can be roughly related to which of the following?
- a) The surface energy liberated during fracture
 - b) The strain energy surrounding dislocations in the glass
 - c) The thermal energy used to produce the glass
 - d) None of the above
3. In an FCC unit cell, how many complete atoms are contributed to the unit cell from the atoms in the corners of the cell?
- a) 8
 - b) 4
 - c) 1
 - d) 2
4. A lump of solid gold having a mass of 38g is placed into a cylinder of water that is already full to the top. How much water spills out of the cylinder? Gold has the FCC crystal structure and

$$n_{\text{FCC}} = 4$$

$$A_{\text{Au}} = 196.67 \text{ g/mol}$$

$$m = 38 \text{ g}$$

$$R_{\text{Au}} = 0.146 \text{ nm}$$

$$\rho = \frac{nA}{V_c N_A}, \quad V_c = a^3, \quad a_{\text{FCC}} = 2\sqrt{2}R$$

$$\rho = \frac{m}{V} \rightarrow V = \frac{m}{\rho} \text{ (} V \text{ is total vol. of gold, not } V_c \text{)}$$

$$V = \frac{m \cdot (2\sqrt{2}R)^3 \cdot N_A}{nA} = \frac{(38) \cdot (2\sqrt{2} \cdot 0.146 \text{ nm})^3 \cdot 6.022 \times 10^{23}}{4 \cdot 196.67}$$

$$V = 2.05 \times 10^{-6} \text{ m}^3 = 2.05 \text{ cm}^3 = 2.05 \text{ mL}$$